Biomedical Data Acquisition Course (BMD351)

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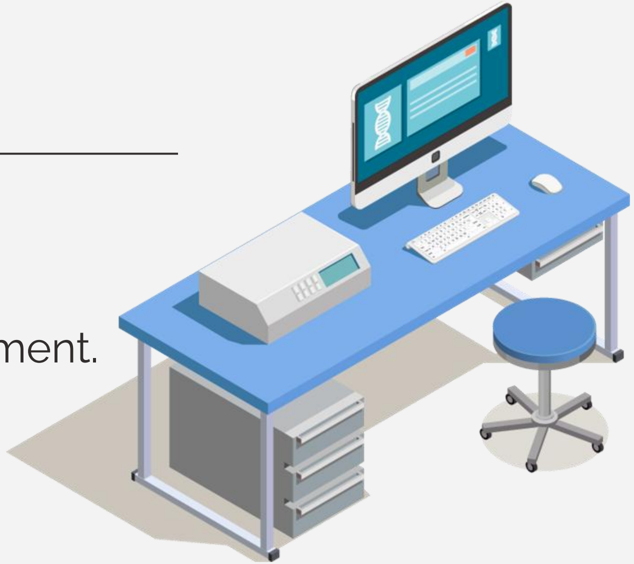


03

Biomedical Imaging

Types of Biomedical Imaging

- X-Ray
- CT scan
- MRI
- Ultrasound
- Microscopy
- Nuclear Medicine
- PET
- SPECT

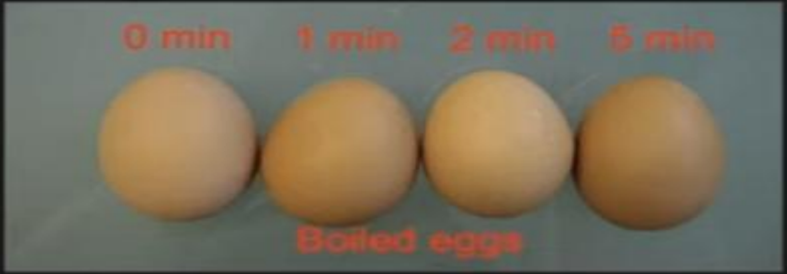
Types of Biomedical Imaging

X-Ray	CT scan	MRI
Ultrasound	Microscopy	Nuclear Medicine

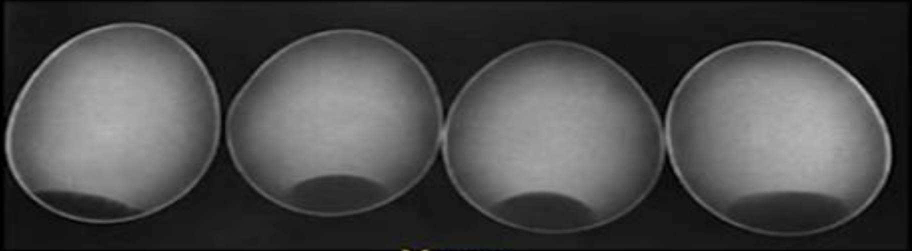
Medical Imaging Modalities

- Medical imaging is the process of visual representation of the structure and function of different tissues and organs of the human body for clinical purposes .
- The purpose of imaging modalities: Diagnosis, Screening, Therapy Planning and Follow up
- Obtain information on:
  - Morphology (organ and cellular level).
  - Function (Physiology, Biochemistry).
  - Bloodflow.
- Examples:
  - X-ray radiography
  - X-ray computed tomography (CT)
  - magnetic resonance imaging (MRI)
  - magnetic resonance spectroscopy (MRS)
  - positron emission tomography (PET)
  - digital mammography
  - single-photon emission computed tomography (SPECT)
  - ultrasonic

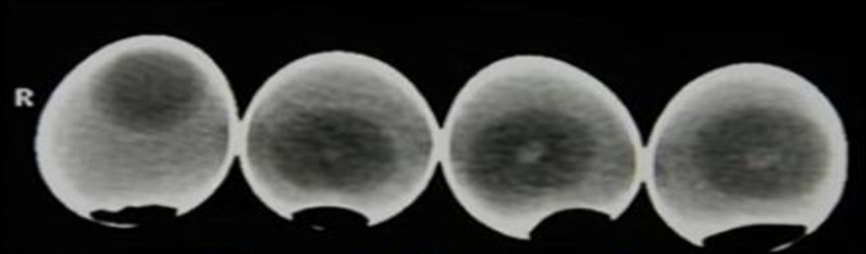
Imaging modalities



Destructive testing

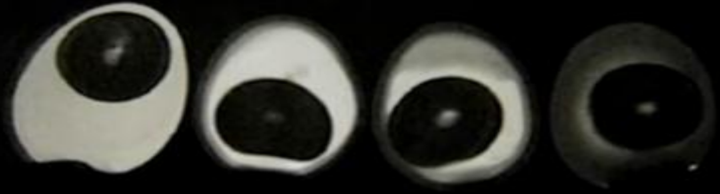


X-ray

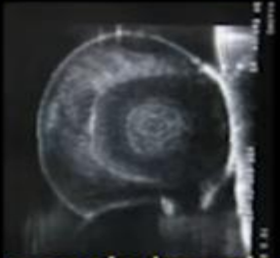


kV 140
mA 159

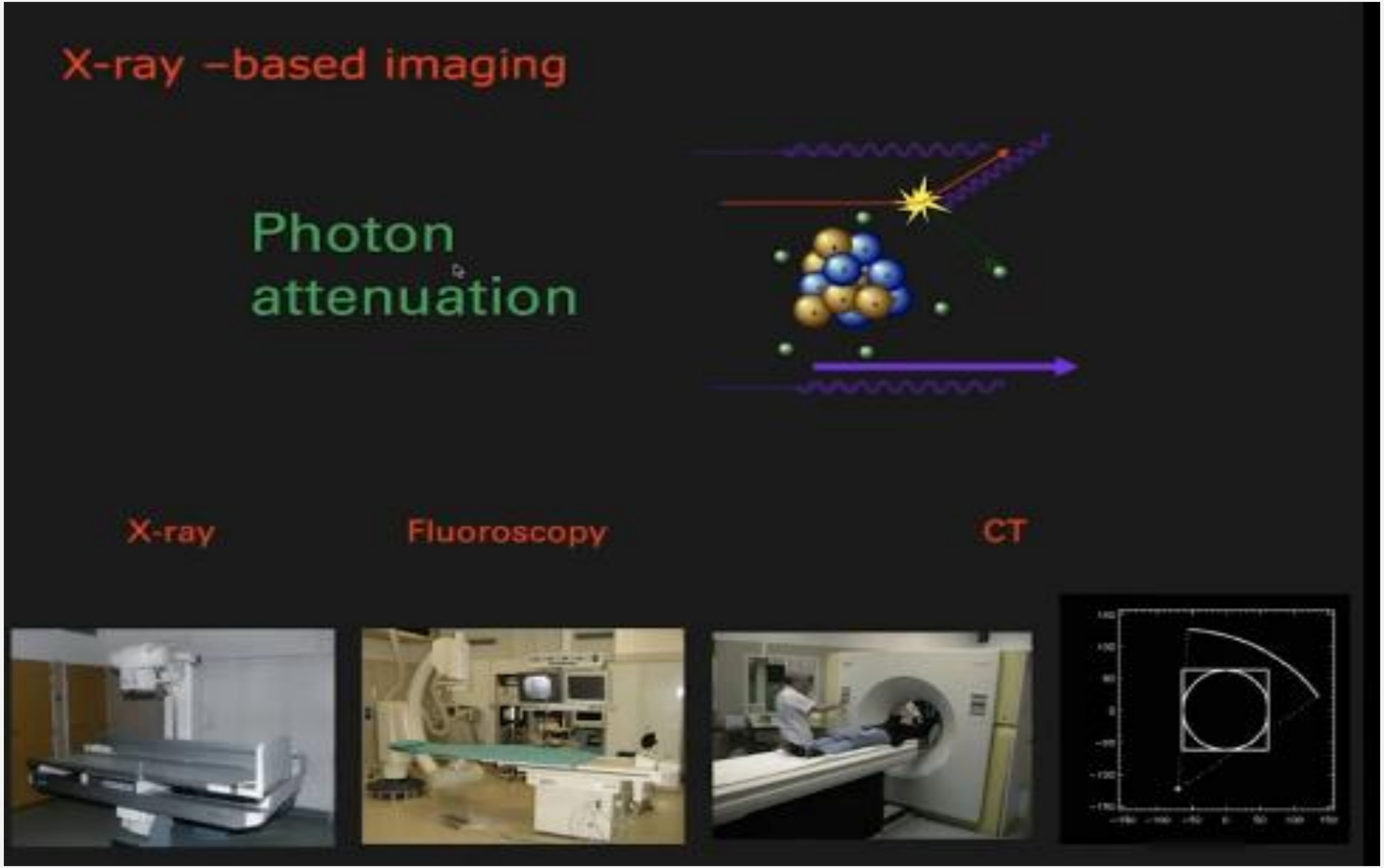
Computed Tomography



Magnetic Resonance Imaging

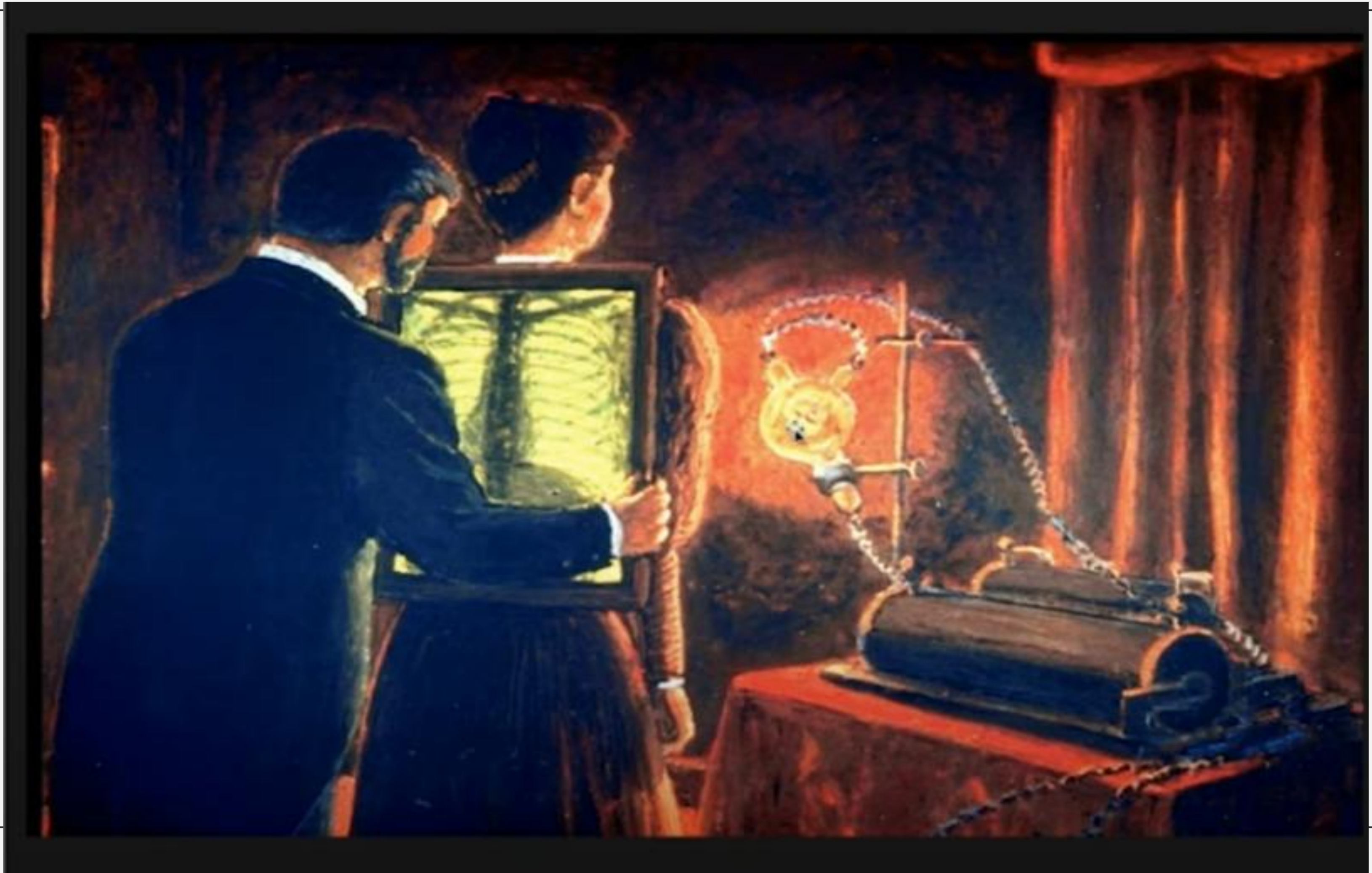


Ultrasonic imaging

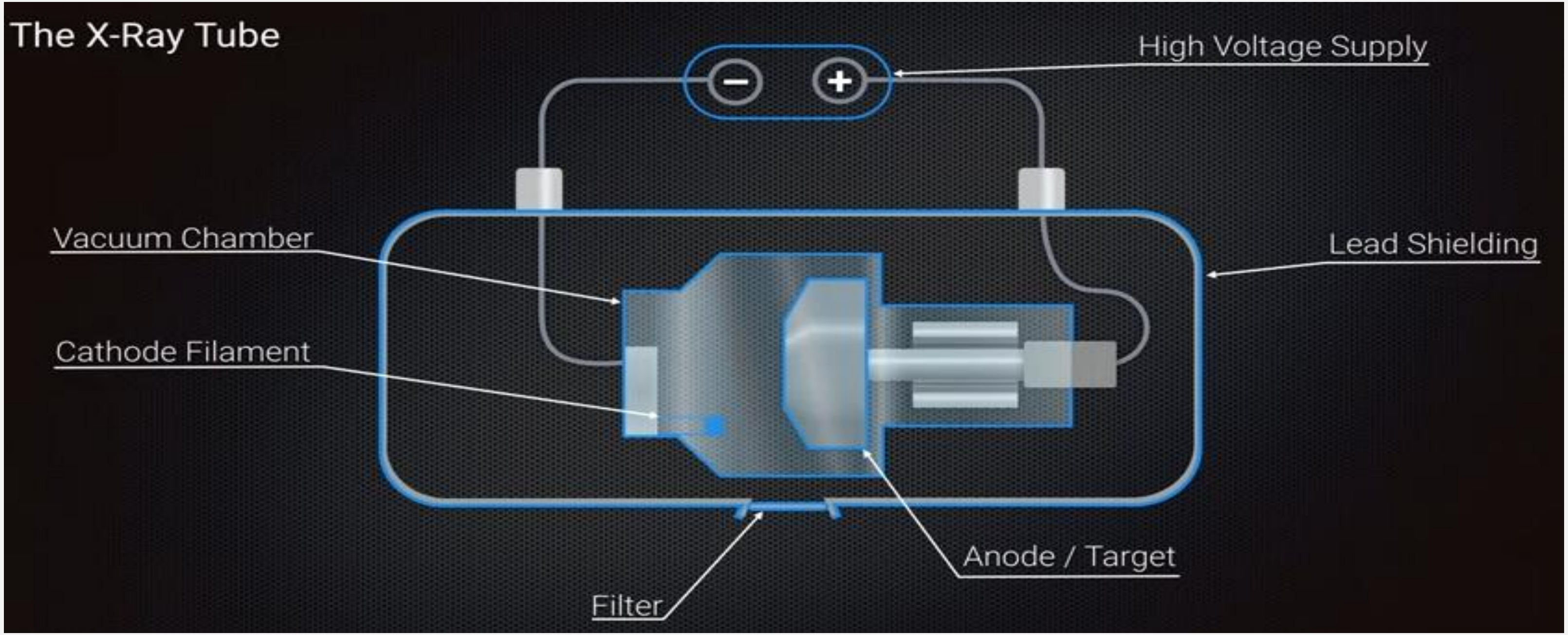

X-ray –based imaging
Photon attenuation
X-ray
Fluoroscopy
CT

X-rays

- Medical imaging techniques are developed after the discovery of X-rays.
- Radiologists gave names to X-ray basis as “X-rays” or “plane film” used for diagnosing bone fractures and chest abnormalities.
- Fluoroscopy was developed due to a more powerful beam of X-ray for diagnosing the patient abnormalities.
- Fluoroscopy is now converted into computed tomography (CT).
- X-rays are a form of electromagnetic radiation, similar to visible light. Unlike light, however, x-rays have higher energy and can pass through most objects, including the body.
- Medical x-rays are used to generate images of tissues and structures inside the body.
- If x-rays traveling through the body also pass through an x-ray detector on the other side of the patient, an image will be formed that represents the “shadows” formed by the objects inside of the body. The x-ray images that result from this process are called radiographs.

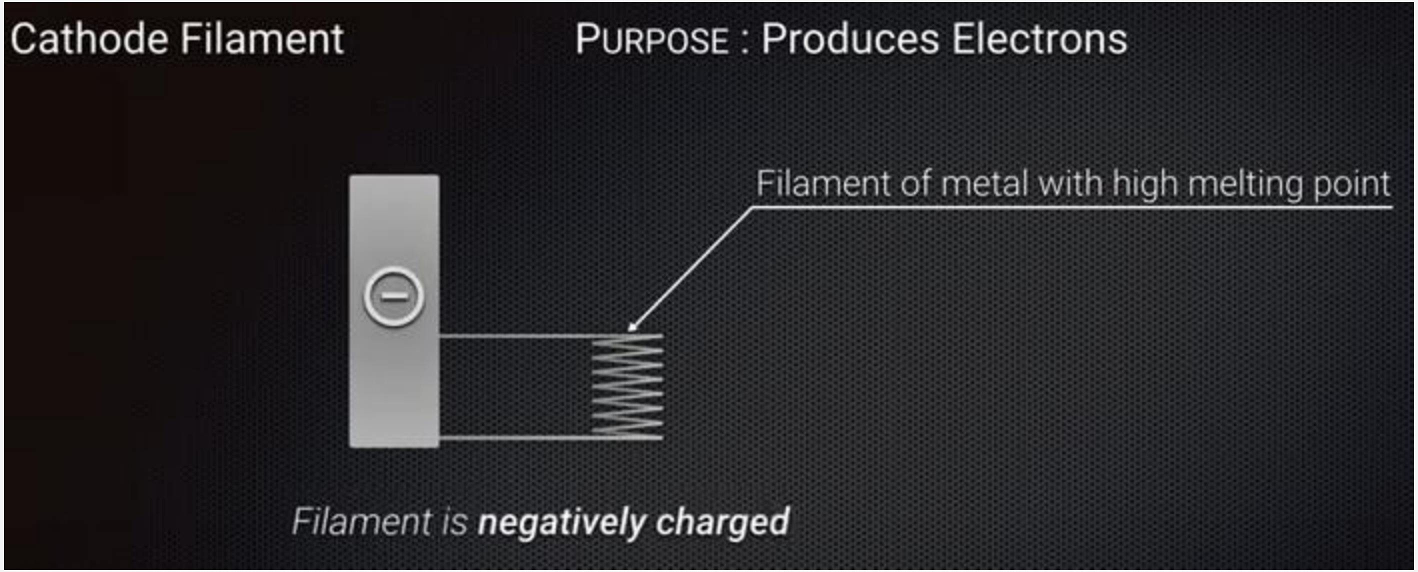


X-ray Tube



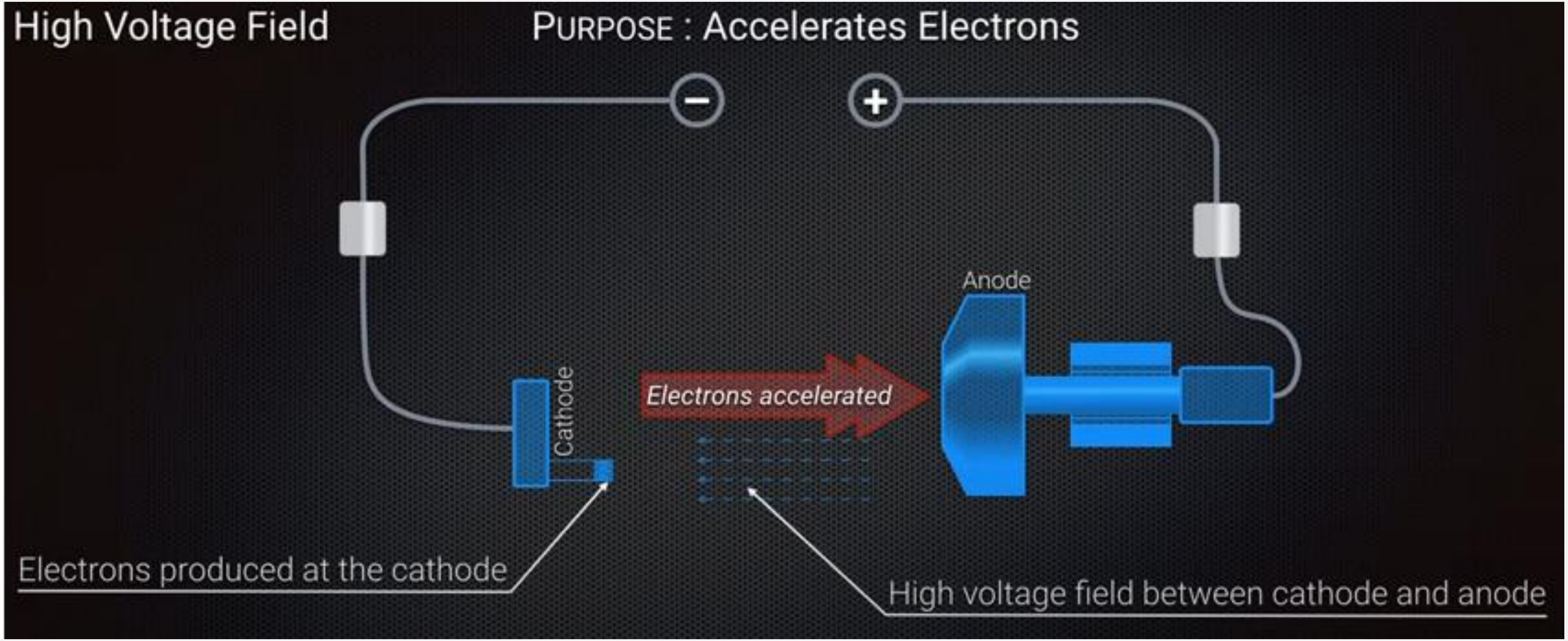
The X-Ray Tube
High Voltage Supply
Vacuum Chamber
Lead Shielding
Cathode Filament
Anode / Target
Filter

X-ray Tube



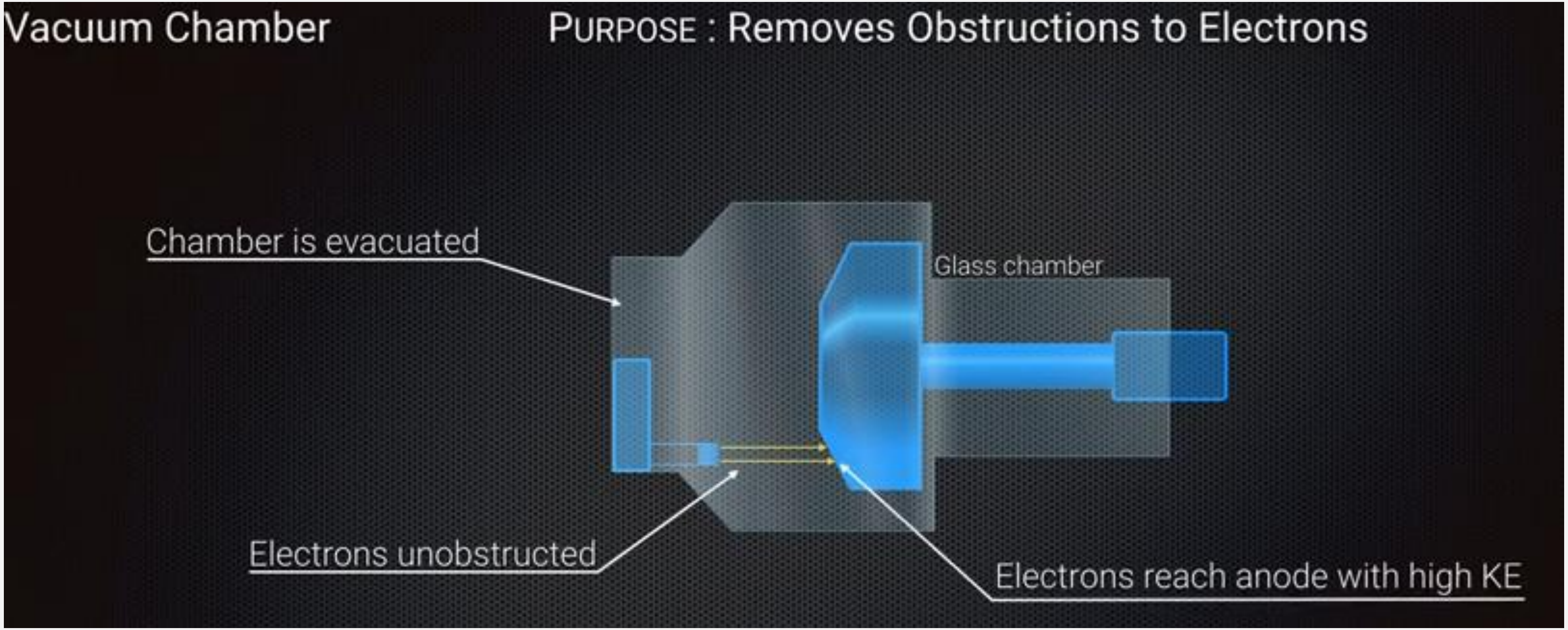
Cathode Filament
PURPOSE : Produces Electrons
Filament of metal with high melting point
Filament is **negatively charged**

X-ray Tube



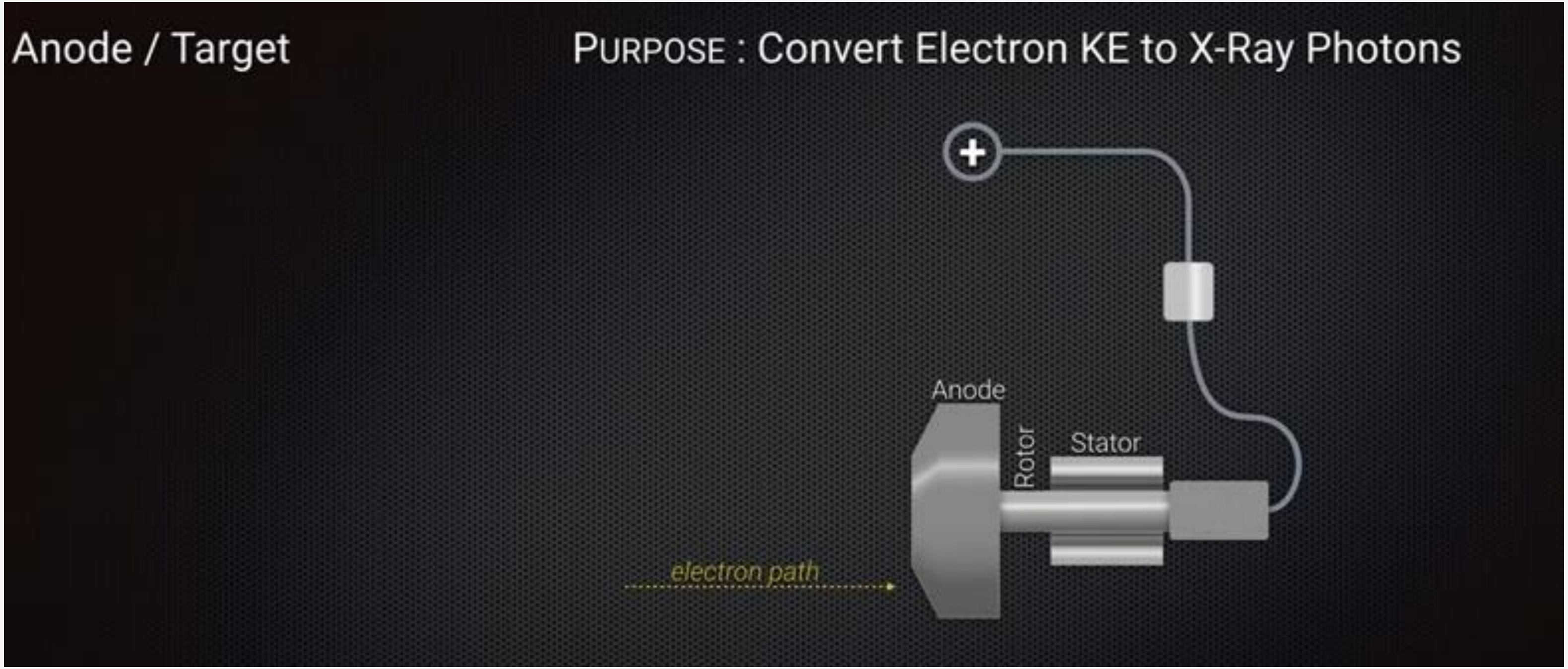
High Voltage Field
PURPOSE : Accelerates Electrons
Anode
Cathode
Electrons accelerated
Electrons produced at the cathode
High voltage field between cathode and anode

X-ray Tube



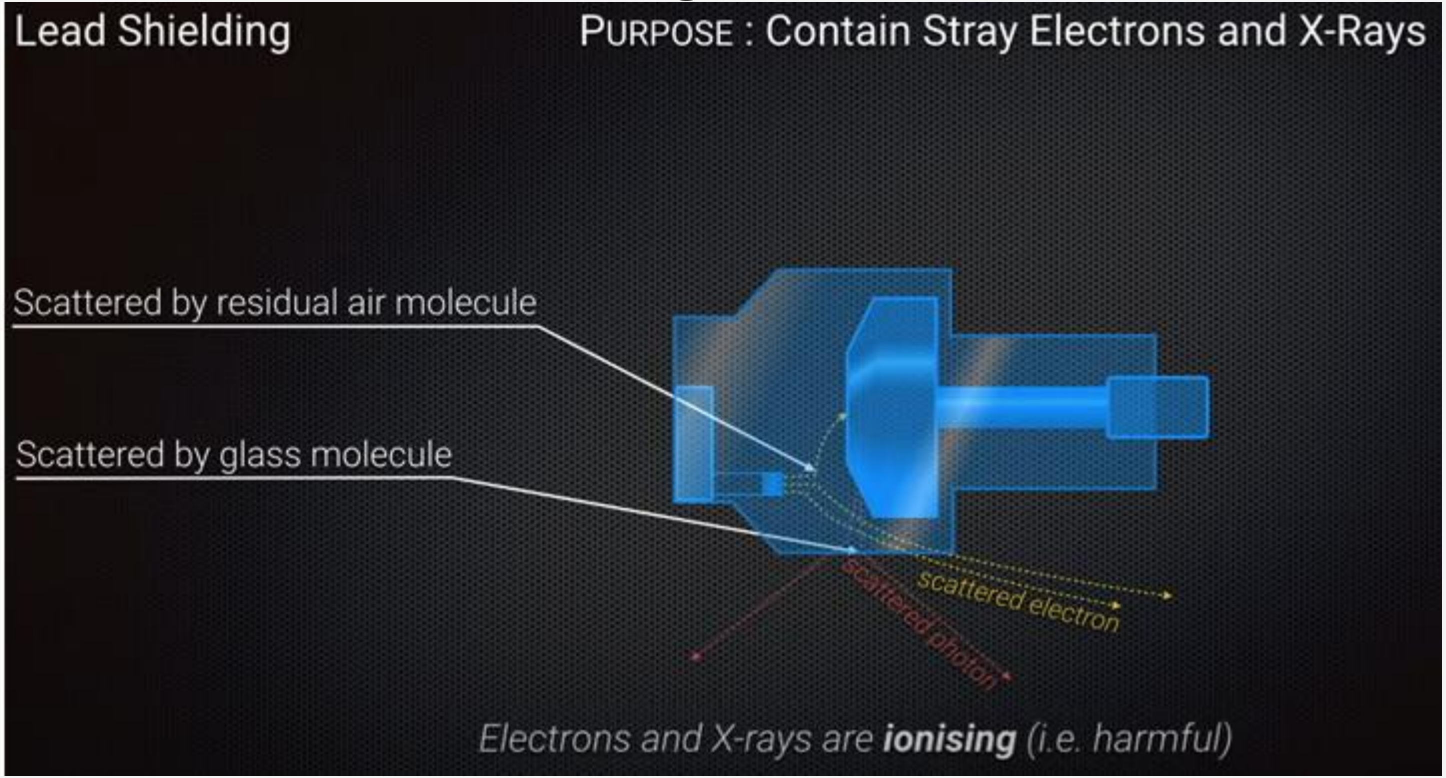
Vacuum Chamber
PURPOSE : Removes Obstructions to Electrons
Chamber is evacuated
Glass chamber
Electrons unobstructed
Electrons reach anode with high KE

X-ray Tube

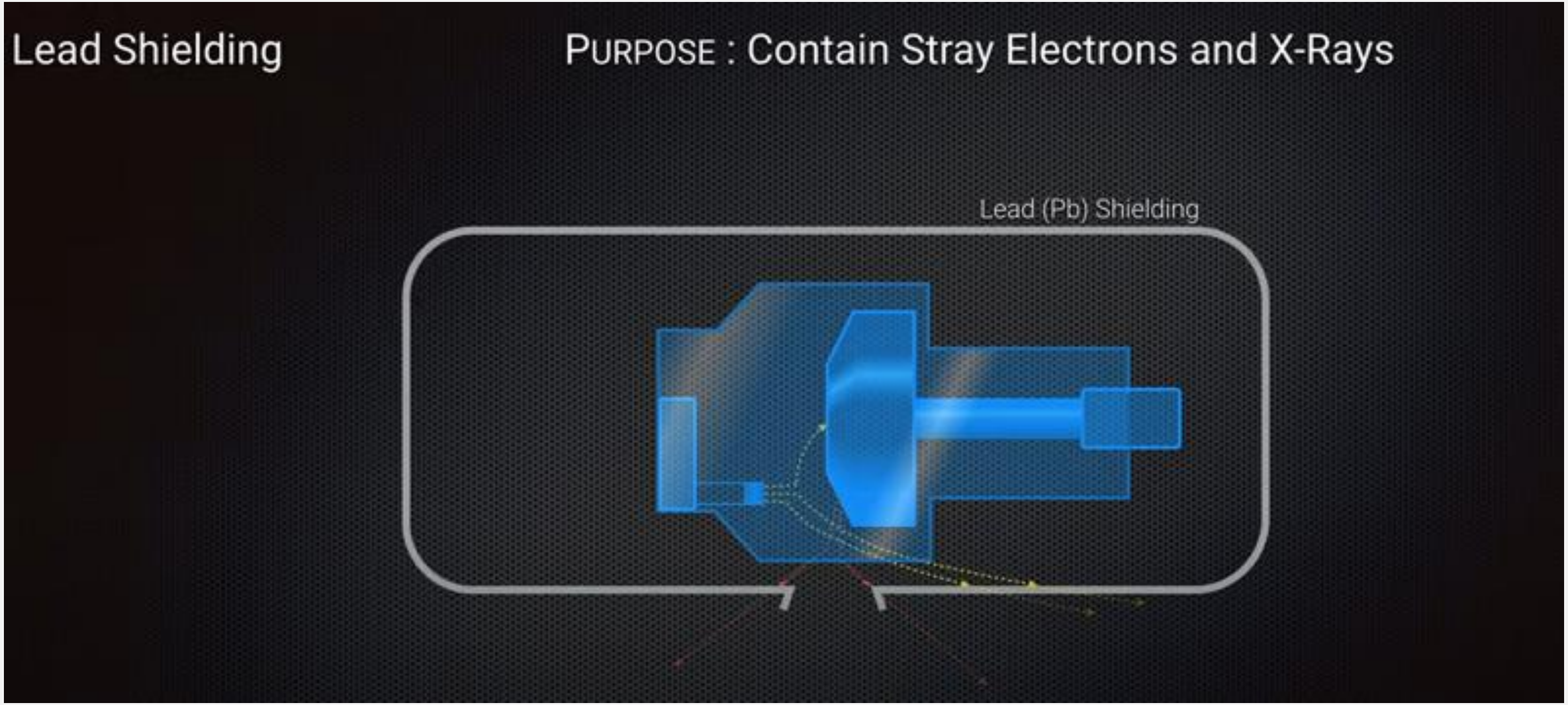


Anode / Target
PURPOSE : Convert Electron KE to X-Ray Photons
Anode
Rotor
Stator
electron path

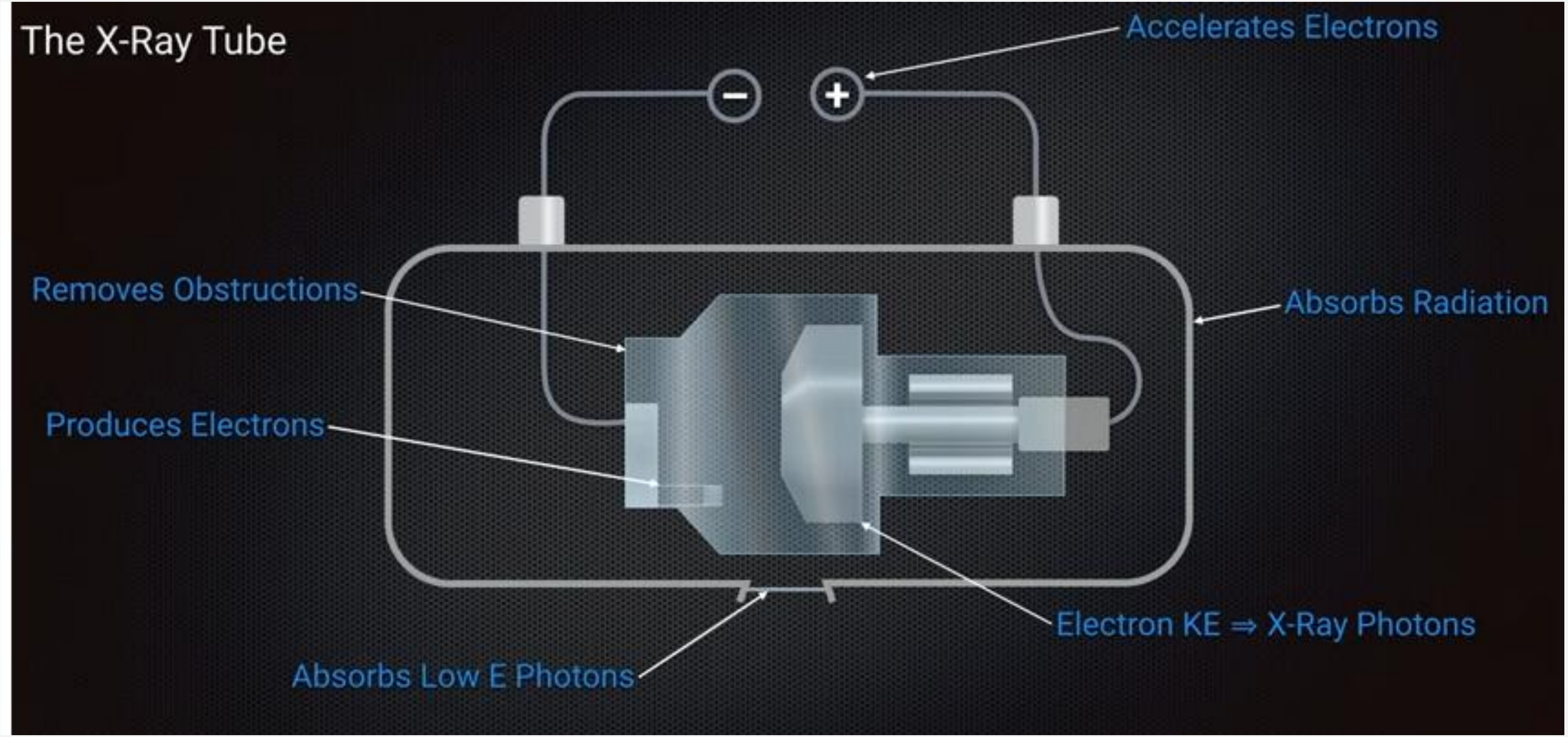
X-ray Tube


Lead Shielding
PURPOSE : Contain Stray Electrons and X-Rays
Scattered by residual air molecule
Scattered by glass molecule
scattered electron
scattered photon
Electrons and X-rays are **ionising** (i.e. harmful)

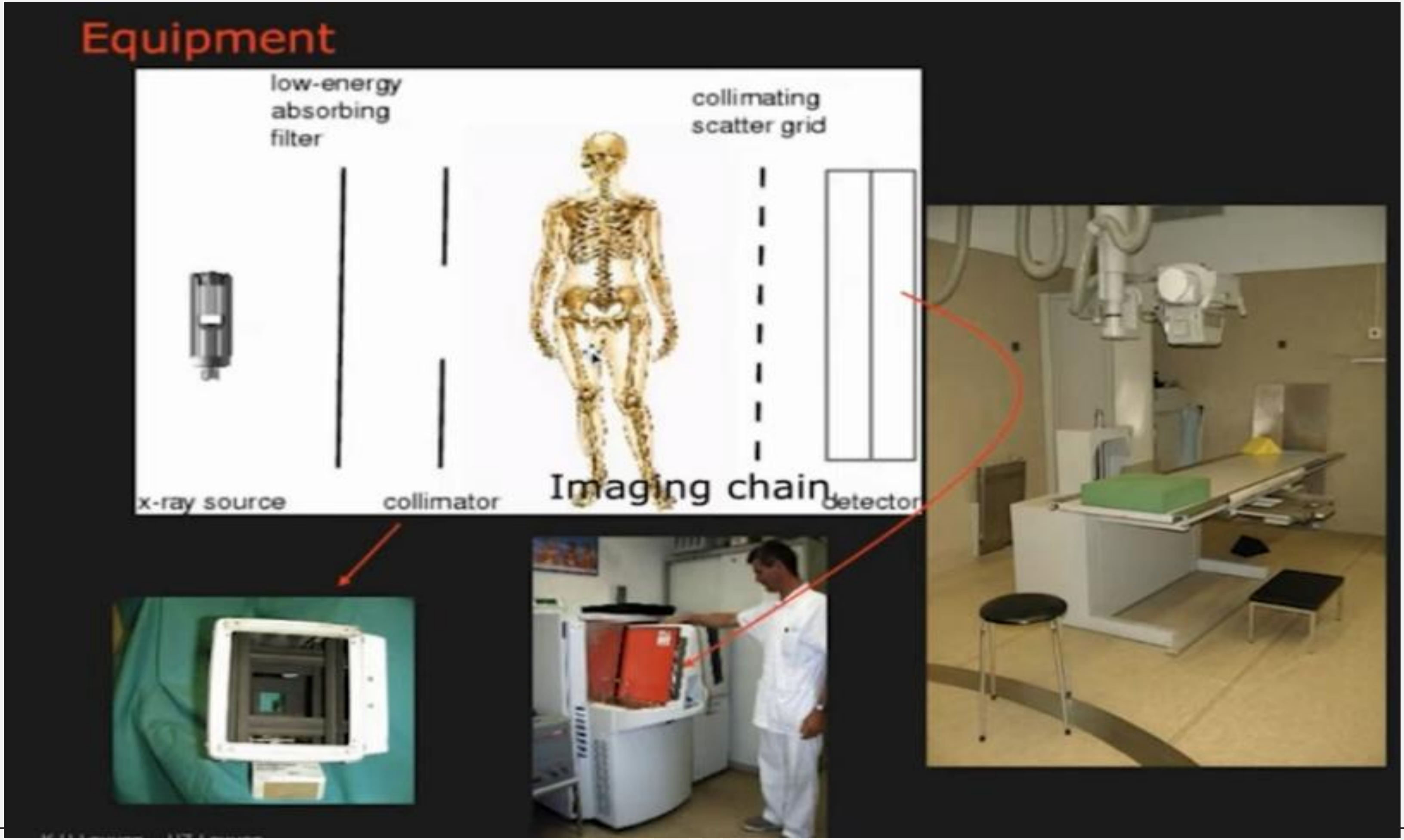
X-ray Tube


Lead Shielding
PURPOSE : Contain Stray Electrons and X-Rays
Lead (Pb) Shielding

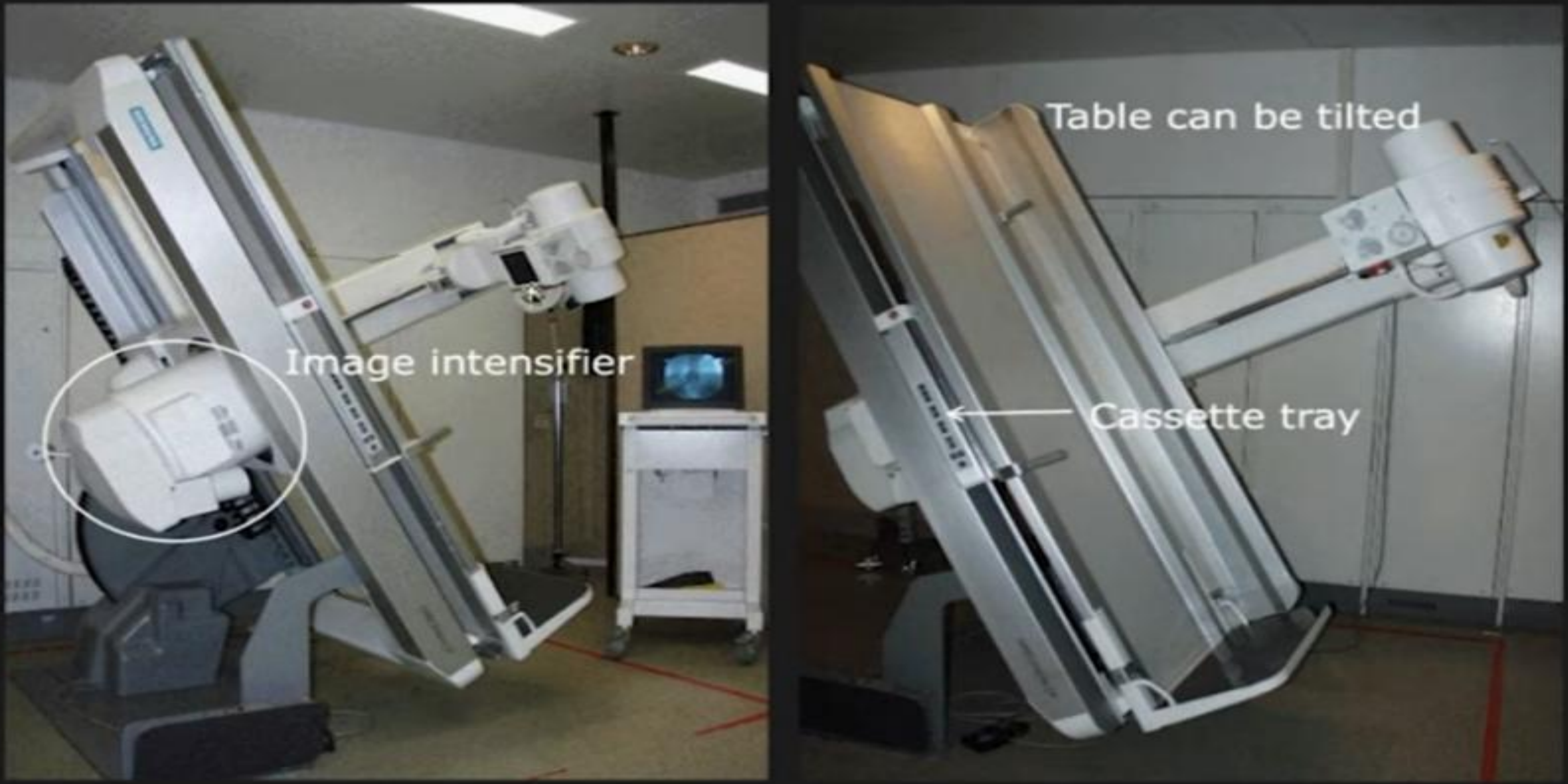
X-rays



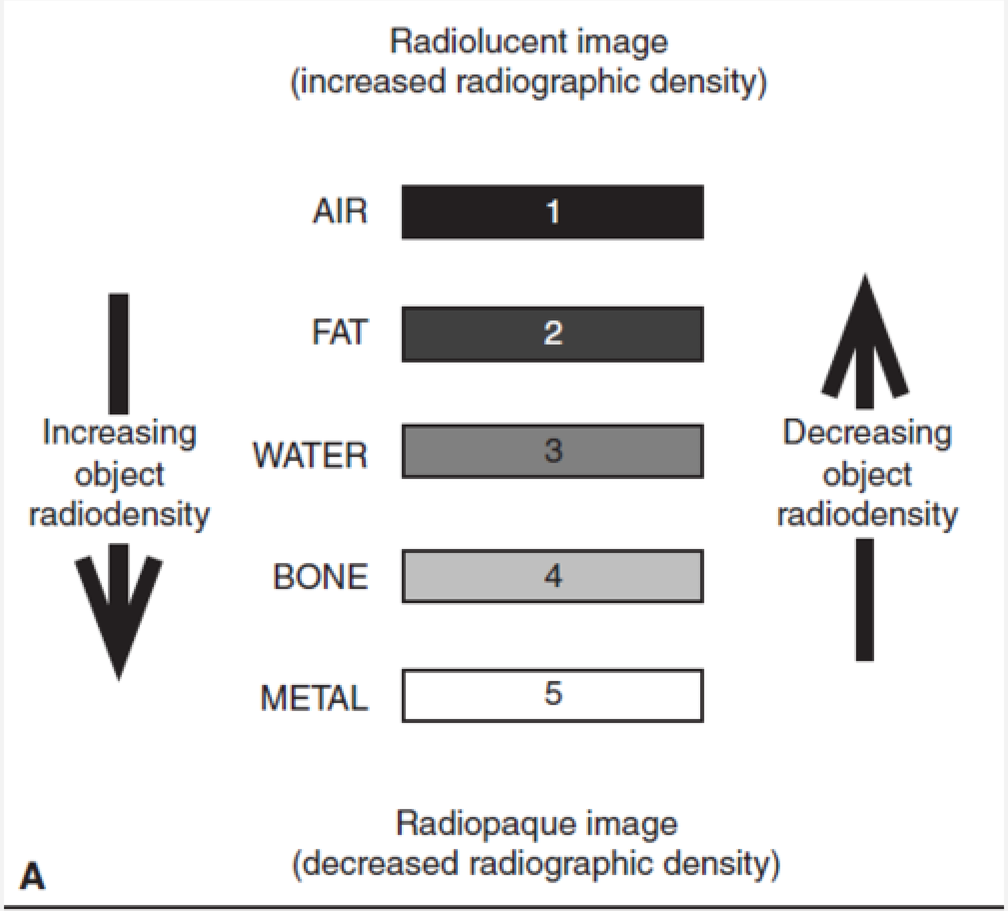
The X-Ray Tube
Accelerates Electrons
Removes Obstructions
Absorbs Radiation
Produces Electrons
Electron KE ⇒ X-Ray Photons
Absorbs Low E Photons

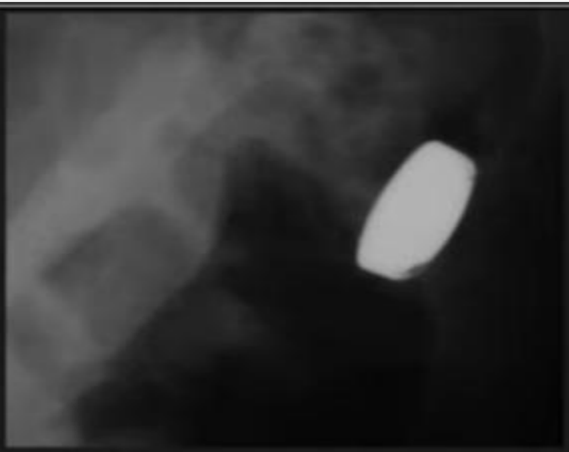

Equipment
low-energy
absorbing
filter
collimating
scatter grid
x-ray source
collimator
Imaging chain
detector

General purpose radiographic room

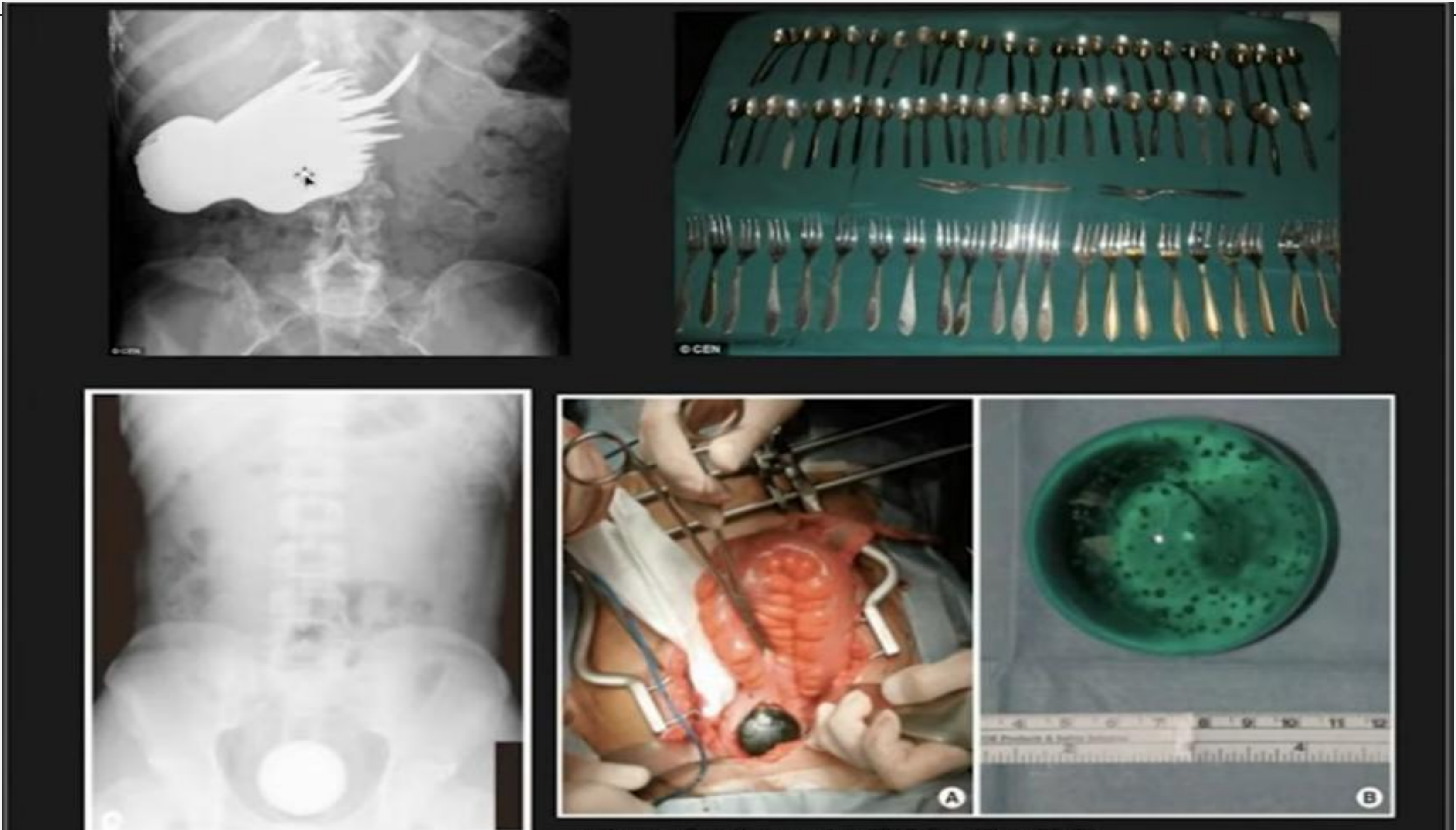


- air absorbs less x-ray less energy , appear black.
- fat = dark gray.
- soft tissue or fluid (heart muscle or blood) gray.
- Calcium = bone = more absorption.
- Metal more and more absorption.





K.U.Leuven – UZ Leuven
Medical Imaging Research Center



J Korean Soc Coloproctol. 2012 Feb;28(1):56-60

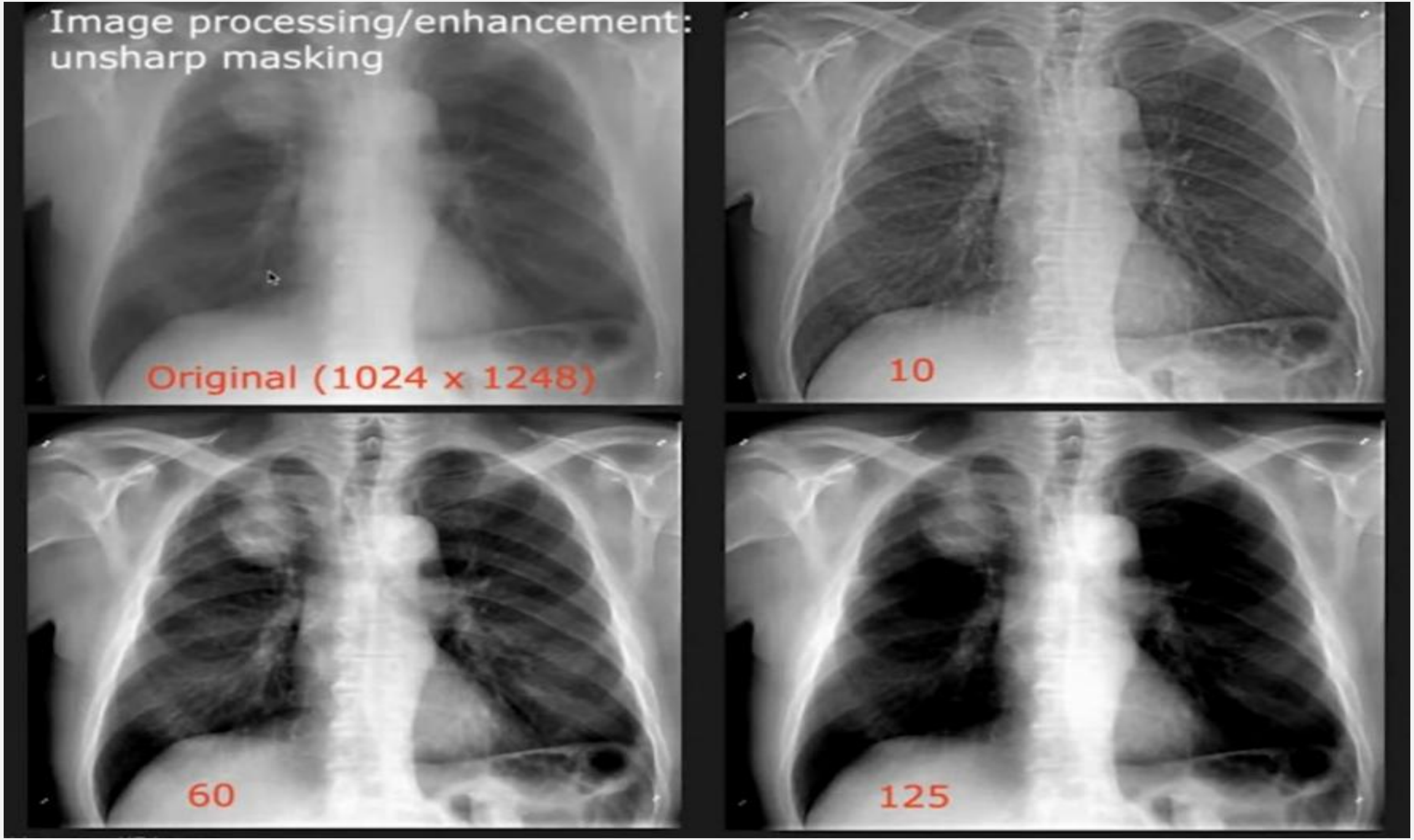
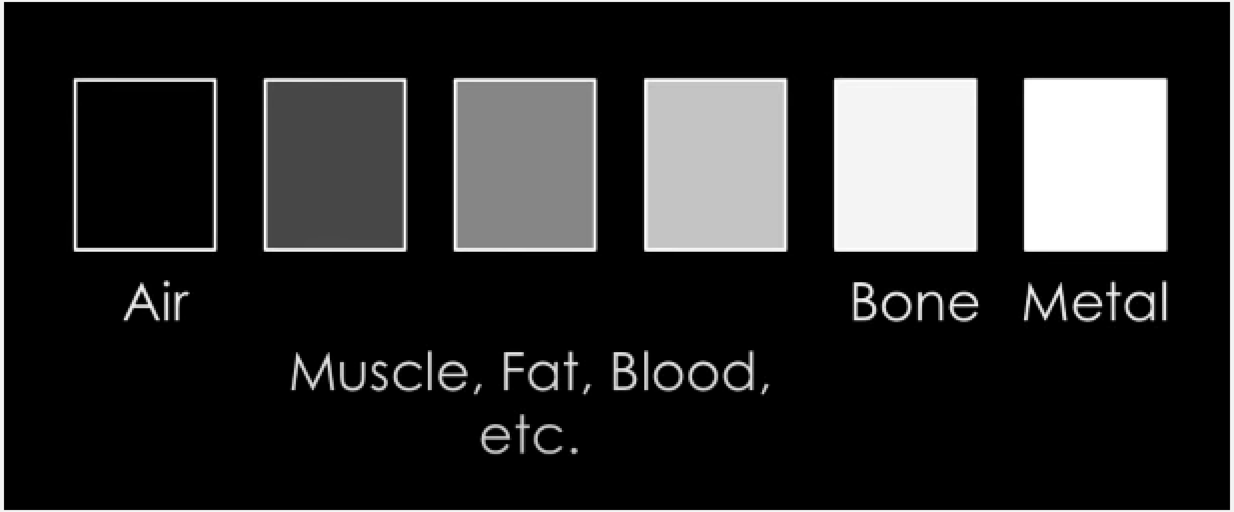


Image processing/enhancement:
unsharp masking
Original (1024 x 1248)
10
60
125


Air
Bone
Metal
Muscle, Fat, Blood,
etc.

Chest X-Ray Images (Pneumonia)

Figure S6. Illustrative Examples of Chest X-Rays in Patients with Pneumonia, Related to Figure 6

The normal chest X-ray (left panel) depicts clear lungs without any areas of abnormal opacification in the image. Bacterial pneumonia (middle) typically exhibits a focal lobar consolidation, in this case in the right upper lobe (white arrows), whereas viral pneumonia (right) manifests with a more diffuse "interstitial" pattern in both lungs.

http://www.cell.com/cell/fulltext/S0092-8674(18)30154-5

- https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia

COVID-19 Chest X-ray dataset

COVID-19 Chest X-ray dataset

Hello guys! This dataset contains the chest X-ray images of Normal people and the COVID-19 affected people. This dataset is the combined and modified version of COVID-19 chest xray and chest-xray-dataset datasets. It can be used to detect COVID-19 patients effectively. I have already published a notebook using this dataset, where I have used a Convolutional Neural Network and achieved a very good result. You can have a look at it. This will help you to gain a better understanding of image classification using CNN.

https://www.kaggle.com/discussions/general/186250

Knee Osteoarthritis Dataset with Severity Grading

Predict the Severity Grading of Knee Osteoarthritis

Data Card Code (22) Discussion (3)

About Dataset

Usability

8.75

License

Attribution 4.0 International (C...

Context

Knee osteoarthritis is defined by degeneration of the knee's articular cartilage the flexible, slippery material that normally protects bones from joint friction and imp
affect nearby soft tissues. Knee osteoart
to as simply knee arthritis. Many other le
pseudogout, and reactive arthritis.

Content

This dataset contains knee X-ray data for both knee joint detection and knee KL grading. The Grade descriptions are as follows:

- Grade 0: Healthy knee image.
- Grade 1 (Doubtful): Doubtful joint narrowing with possible osteophytic lipping
- Grade 2 (Minimal): Definite presence of osteophytes and possible joint space narrowing
- Grade 3 (Moderate): Multiple osteophytes, definite joint space narrowing, with mild sclerosis.
- Grade 4 (Severe): Large osteophytes, significant joint narrowing, and severe sclerosis.

Thank you!